

New Wheels Project

Introduction to SQL – Aryaman Modi

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
-- Q1.a
```

```
SELECT COUNT(DISTINCT customer_id) AS total_customers_who_placed_orders  
FROM order_t;
```

```
-- Q1.b
```

```
SELECT  
    c.state,  
    COUNT(DISTINCT o.customer_id) AS customers_in_state  
FROM order_t o  
JOIN customer_t c  
    ON o.customer_id = c.customer_id  
GROUP BY c.state  
ORDER BY customers_in_state DESC;
```

Output:

Test Cases Run SQL

Result: **Passed**

✓ Query 1

Query:

```
SELECT COUNT(DISTINCT customer_id) AS total_customers_who_placed_orders
FROM order_t
```

Output:

Showing 1 rows

total_customers_who_...
994

✓ Query 2

Query:

```
SELECT
  c.state,
  COUNT(DISTINCT o.customer_id) AS customers_in_state
FROM order_t o
JOIN customer_t c
  ON o.customer_id = c.customer_id
GROUP BY c.state
ORDER BY customers_in_state DESC
```

Output:

Showing first 10 rows out of 49 rows

state	customers_in_state
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29

Observations and Insights:

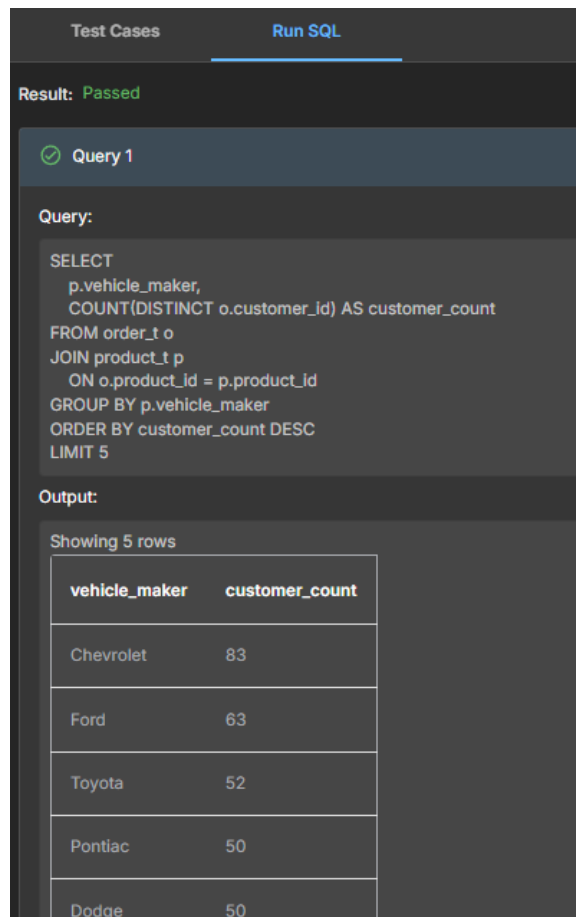
- A total of **994 customers** placed orders, showing strong platform usage.
- **Texas and California (97 each)**, followed by **Florida (86)** and **New York (69)**, contribute the largest share of customers, making them key focus markets.
- Since most customers come from a few states, improving service quality and delivery experience in these regions can significantly boost satisfaction and repeat business.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN product_t p
    ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
```

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result status is 'Result: Passed'. A green checkmark icon is next to 'Query 1'. The query text is displayed in a code block, and the output is shown as a table with 5 rows.

Result: Passed

Query 1

Query:

```
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN product_t p
    ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- The top preferred makers are Chevrolet (83 customers), followed by Ford (63) and Toyota (52), showing strong demand for these brands.
- These makers attract the highest unique customer interest, so increasing their inventory can boost sales.
- Targeted promotions on these brands may improve conversions and meet customer expectations more effectively.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, cust_count
FROM (
  WITH maker_pref AS (
    SELECT
      c.state,
      p.vehicle_maker,
      COUNT(DISTINCT o.customer_id) AS cust_count,
      RANK() OVER (
        PARTITION BY c.state
        ORDER BY COUNT(DISTINCT o.customer_id) DESC
      ) AS rnk
    FROM order_t o
    JOIN customer_t c
      ON o.customer_id = c.customer_id
    JOIN product_t p
      ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
  )
  SELECT * FROM maker_pref WHERE rnk = 1
)
ORDER BY state
```

Output:

Test Cases
Run SQL

Result: Passed

✓ Query 1

Query:

```

SELECT state, vehicle_maker, cust_count
FROM (
  WITH maker_pref AS (
    SELECT
      c.state,
      p.vehicle_maker,
      COUNT(DISTINCT o.customer_id) AS cust_count,
      RANK() OVER (
        PARTITION BY c.state
        ORDER BY COUNT(DISTINCT o.customer_id) DESC
      ) AS mk
    FROM order_t o
    JOIN customer_t c
      ON o.customer_id = c.customer_id
    JOIN product_t p
      ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
  )
  SELECT * FROM maker_pref WHERE mk = 1
)
ORDER BY state

```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	cust_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
Arkansas	Chevrolet	1

Observations and Insights:

- Each state shows a clear favorite maker (e.g., Alabama – Dodge (5), Alaska – Chevrolet (2), Arizona – Pontiac / Cadillac (tied 3)), indicating strong regional brand preferences.
- Ties exist in some states (equal cust_count for multiple makers), so inventory should include the top few brands rather than only the single leader.
- Action: align regional inventory and targeted promotions to these state-level preferences to reduce shipping time, increase conversion, and improve after-sales satisfaction.

Question 4: Find the overall average rating given by the customers.

What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

-- Q4

DROP TABLE IF EXISTS temp_rating_map;

CREATE TEMP TABLE temp_rating_map AS

SELECT

 quarter_number,

 CASE customer_feedback

 WHEN 'Very Bad' THEN 1

 WHEN 'Bad' THEN 2

 WHEN 'Okay' THEN 3

 WHEN 'Good' THEN 4

 WHEN 'Very Good' THEN 5

 END AS rating_value

FROM order_t;

SELECT ROUND(AVG(rating_value), 2) AS overall_avg_rating

FROM temp_rating_map;

SELECT

 quarter_number,

 ROUND(AVG(rating_value), 2) AS avg_rating_per_quarter

FROM temp_rating_map

GROUP BY quarter_number

ORDER BY quarter_number

LIMIT 50;

Output:

Test Cases
Run SQL

Result: Passed

Query 1

Query 2

Query 3

Query:

```
SELECT ROUND(AVG(rating_value), 2) AS overall_avg_rating
FROM temp_rating_map
```

Output:

Showing 1 rows

overall_avg_rating
3.14

Query 4

Query:

```
SELECT
  quarter_number,
  ROUND(AVG(rating_value), 2) AS avg_rating_per_quarter
FROM temp_rating_map
GROUP BY quarter_number
ORDER BY quarter_number
LIMIT 50
```

Output:

Showing 4 rows

quarter_number	avg_rating_per_quarter
1	3.55
2	3.35
3	2.96
4	2.4

Observations and Insights:

- The overall average rating is 3.14, showing that customer satisfaction is slightly above neutral.
- Ratings drop steadily from Q1 (3.55) to Q4 (2.40), indicating customers are becoming increasingly dissatisfied over time.
- This downward trend suggests worsening after-sales service or shipping experience that the business needs to address urgently.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT  
  
    quarter_number,  
  
  
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END)  
        / COUNT(*), 2) AS pct_very_bad,  
  
  
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END)  
        / COUNT(*), 2) AS pct_bad,  
  
  
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END)  
        / COUNT(*), 2) AS pct_okay,  
  
  
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END)  
        / COUNT(*), 2) AS pct_good,  
  
  
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END)  
        / COUNT(*), 2) AS pct_very_good  
  
  
FROM order_t  
  
GROUP BY quarter_number  
  
ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
  quarter_number,

  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END)
    / COUNT(*), 2) AS pct_very_bad,

  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END)
    / COUNT(*), 2) AS pct_bad,

  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END)
    / COUNT(*), 2) AS pct_okay,

  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END)
    / COUNT(*), 2) AS pct_good,

  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END)
    / COUNT(*), 2) AS pct_very_good

FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	pct_very_bad	pct_bad	pct_okay	pct_good	pct_very_good
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

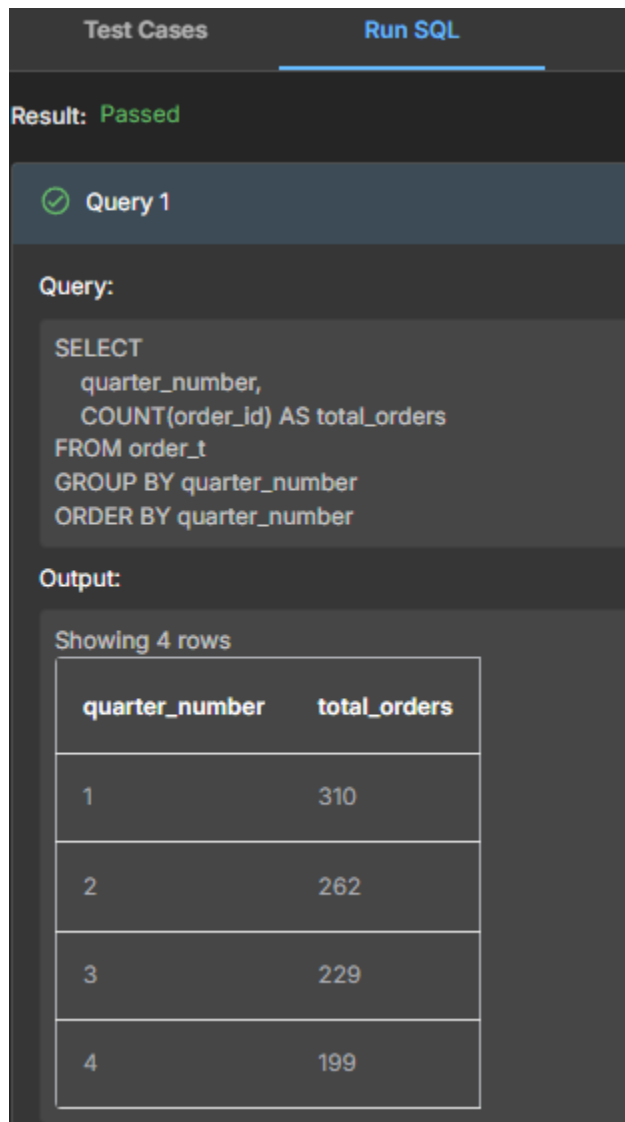
- The share of negative feedback (Very Bad + Bad) rises sharply from ~22% in Q1 to ~60% in Q4, while positive feedback (Good + Very Good) drops from ~58% to ~20%.
- This indicates customers are becoming significantly more dissatisfied over time, especially in the last two quarters.
- The trend suggests worsening issues in after-sales support or delivery experience that New-Wheels needs to address urgently.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL execution environment with two tabs: 'Test Cases' and 'Run SQL'. The 'Run SQL' tab is active, and the result is 'Passed'. Below this, 'Query 1' is listed with a green checkmark. The query text is displayed in a dark box, and the output is shown as a table with 4 rows.

Result: **Passed**

Query 1

Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations and Insights:

- Quarterly orders show a steady decline from 310 orders in Q1 to 199 orders in Q4, indicating weakening customer demand over time.
- This downward trend aligns with the falling customer satisfaction seen in earlier questions.
- The drop suggests underlying issues in service quality or customer experience that may be discouraging repeat purchases.

Question 7: Calculate the net revenue generated by the company.
What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
DROP TABLE IF EXISTS temp_revenue;
```

```
CREATE TEMP TABLE temp_revenue AS
```

```
SELECT
```

```
    quarter_number,
```

```
    SUM( quantity * (vehicle_price * (1 - discount/100.0)) ) AS net_revenue
```

```
FROM order_t
```

```
GROUP BY quarter_number;
```

```
SELECT
```

```
    quarter_number,
```

```
    ROUND(net_revenue, 2) AS net_revenue,
```

```
    ROUND(
```

```
        (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
```

```
        / LAG(net_revenue) OVER (ORDER BY quarter_number) * 100
```

```
    , 2) AS qoq_percent_change
```

```
FROM temp_revenue
```

```
ORDER BY quarter_number
```

```
LIMIT 50;
```

Output:

Test Cases
Run SQL

✓ Query 2

✓ Query 3

Query:

```

SELECT
  quarter_number,
  ROUND(net_revenue, 2) AS net_revenue,
  ROUND(
    (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
    / LAG(net_revenue) OVER (ORDER BY quarter_number) * 100
    , 2) AS qoq_percent_change
FROM temp_revenue
ORDER BY quarter_number
LIMIT 50

```

Output:

Showing 4 rows

quarter_number	net_revenue	qoq_percent_change
1	39421580.16	
2	32715830.34	-17.01
3	29229896.19	-10.66
4	23346779.63	-20.13

Observations and Insights:

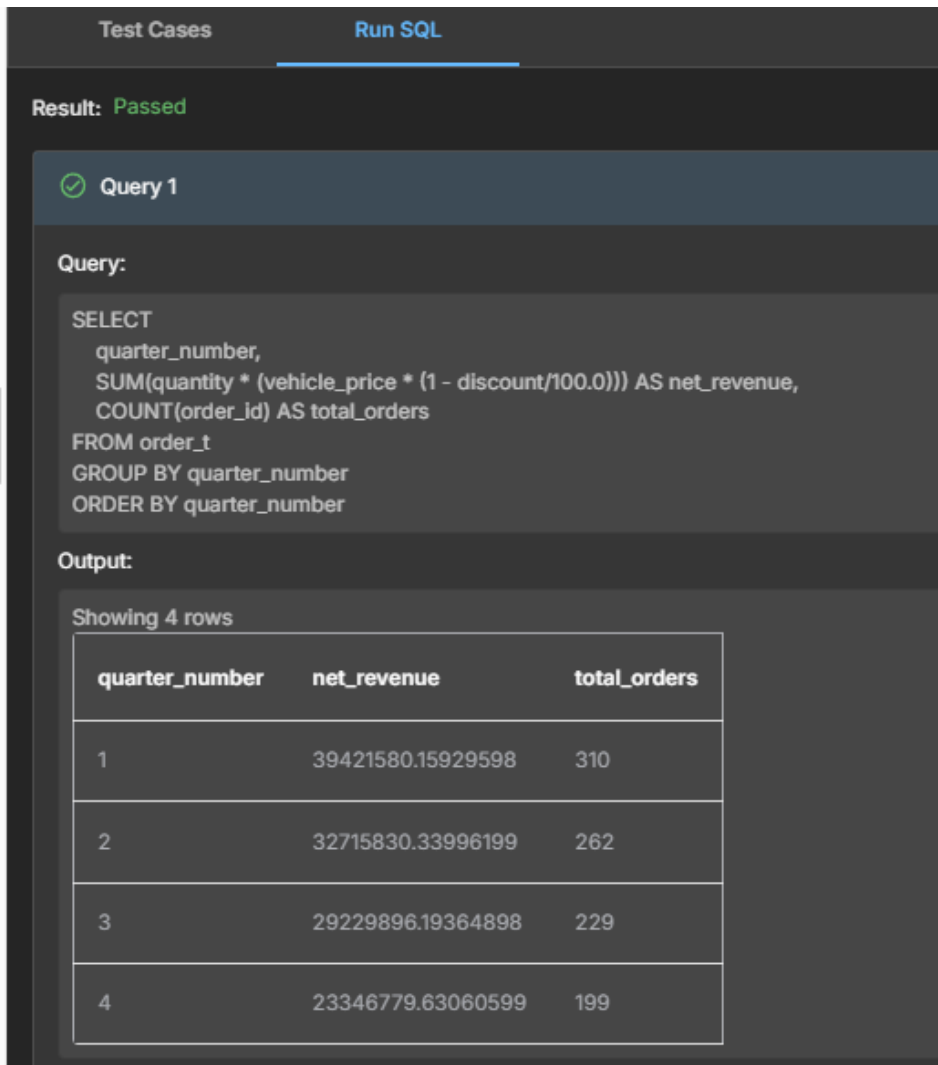
- Net revenue drops consistently from ₹39.4M in Q1 to ₹23.3M in Q4, showing a major decline in business performance.
- QoQ revenue falls sharply: $-17\% \rightarrow -10.66\% \rightarrow -20.13\%$, indicating accelerating loss of sales momentum.
- This strong downward trend aligns with earlier findings (falling ratings, rising negative feedback), suggesting deteriorating customer experience is directly impacting revenue.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT  
  
    quarter_number,  
  
    SUM(quantity * (vehicle_price * (1 - discount/100.0))) AS net_revenue,  
  
    COUNT(order_id) AS total_orders  
  
FROM order_t  
  
GROUP BY quarter_number  
  
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL query execution interface. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result is 'Passed'. The query is labeled 'Query 1' and is displayed in a code editor. The output is shown as a table with 4 rows.

Result: **Passed**

Query: **Query 1**

```
SELECT  
    quarter_number,  
    SUM(quantity * (vehicle_price * (1 - discount/100.0))) AS net_revenue,  
    COUNT(order_id) AS total_orders  
FROM order_t  
GROUP BY quarter_number  
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	total_orders
1	39421580.15929598	310
2	32715830.33996199	262
3	29229896.19364898	229
4	23346779.63060599	199

Observations and Insights:

- Both net revenue and order volume decline steadily from Q1 to Q4, confirming a consistent downward business trend.
- The simultaneous drop in revenue and orders suggests falling customer interest, likely driven by poor after-sales experience and increasing negative feedback.
- Immediate action is needed to improve customer satisfaction to stop this decline.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount), 2) AS avg_discount
FROM order_t o
JOIN customer_t c
    ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount DESC;
```

Output:

Test Cases
Run SQL

Result: Passed

✓ Query 1

Query:

```

SELECT
  c.credit_card_type,
  ROUND(AVG(o.discount), 2) AS avg_discount
FROM order_t o
JOIN customer_t c
  ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount DESC

```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	avg_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62
china-unionpay	0.62
americanexpress	0.62
switch	0.61
jcb	0.61
diners-club-us-ca	0.61

Observations and Insights:

- Average discounts vary slightly across card types, with Laser (0.64) and Mastercard (0.63) offering the highest average discounts.
- Most card types cluster around 0.61–0.62, indicating a fairly consistent discounting policy across payment methods.
- Since differences are small, discounts likely have minimal impact on customer choice of payment method.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT  
  
    quarter_number,  
  
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_days_to_ship  
  
FROM order_t  
  
WHERE ship_date IS NOT NULL  
  
GROUP BY quarter_number  
  
ORDER BY quarter_number;
```

Output:

Test Cases

Run SQL

Result: Passed

✓ Query 1

Query:

```
SELECT  
    quarter_number,  
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_days_to_ship  
FROM order_t  
WHERE ship_date IS NOT NULL  
GROUP BY quarter_number  
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_days_to_ship
1	57.17
2	71.11
3	117.76
4	174.1

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
12,47,14,086.32	1000	994	3.14
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
2,33,46,779.63	199	90	29.10%

Business Recommendations

- Improve after-sales support and delivery issues, as negative feedback and ratings have risen sharply, hurting repeat purchases.
- Focus on key markets (Texas, California, Florida, New York) with better inventory planning and regional shipping partners to reduce delays.
- Strengthen customer experience efforts — quicker shipping, better communication, and quality checks — to reverse declining revenue and boost satisfaction.